TECHNICAL PRODUCT BRIEF

EDM

SpeedCloud by NxtGen

Enterprise-grade public cloud built for performance, sovereignty, and scale.



Supplied product visual: SpeedCloud experience and benchmark snapshot.

A practical guide to the platform's technical foundation, service choices, and common workload patterns.

01 | AT A GLANCE

What SpeedCloud is

SpeedCloud is presented in the supplied material as a sovereign, enterprise-grade public cloud platform. Its operating model combines self-service provisioning with managed infrastructure and enterprise support, so teams can move quickly without taking on every layer of platform operations themselves.

The platform is positioned for organizations that need cloud agility while retaining control over data location, operational governance, security, and cost. It is built on open-source technologies, with OpenStack supporting cloud infrastructure and OpenShift supporting cloud-native application platforms.

The most useful way to evaluate SpeedCloud is by workload and operating requirement: choose the infrastructure primitives for virtual machines and networks, use the platform layer for containers and application delivery, and add security, backup, bare metal, or managed services where the workload needs them.

Key takeaways

- Sovereignty: the supplied collateral emphasizes keeping data, operations, and technology within India.
- Open architecture: OpenStack and OpenShift are identified as the core open-source technologies.
- Self-service operations: teams can provision, manage, and scale resources through an intuitive interface.
- Enterprise controls: firewalls, load balancing, encryption, SSL-VPN, and multi-factor authentication are called out.
- Operational support: managed virtual machines, networking, backup, and infrastructure services are available.

Audience and scope

This brief is intended for technology leaders, infrastructure teams, security teams, and application owners evaluating a public-cloud landing zone, a hybrid extension of an existing data center, or a platform for cloud-native services.

02 | TECHNICAL FOUNDATION

How the platform fits together

The platform can be understood as a set of cooperating layers. Each layer answers a different operational question: where workloads run, how they connect, how applications are delivered, and how the environment is secured and operated.

Layer	What it provides	When it is useful
Infrastructure	OpenStack-based compute, networking, virtual machines, storage, and related infrastructure services.	Run enterprise applications, databases, middleware, and development environments without owning physical infrastructure.
Application platform	OpenShift-based container platform for packaging, deploying, scaling, and operating cloud-native applications.	Modernize applications, standardize delivery pipelines, and operate microservices or APIs consistently.
Connectivity and security	Firewalls, advanced load balancing, encryption, SSL-VPN, and multi-factor authentication are identified in the supplied collateral.	Protect ingress and east-west traffic, provide controlled administrator access, and apply identity and encryption controls.
Operations	Self-service provisioning plus managed virtual machines, networking, backup, and infrastructure support.	Reduce day-to-day operational overhead while retaining a clear service boundary and support model.
High-performance compute	BareMetal servers with optional NVIDIA InfiniBand support of up to 400 Gbps per adapter, as stated in the source material.	Large-scale model training, high-performance computing, and workloads that need low-latency, high-throughput interconnects.

A simple operating model

A typical team starts with a project or environment boundary, provisions compute and networking, applies identity and security controls, deploys the application or virtual machines, and then scales or protects the environment with backup and managed services. The exact service names and quotas should be confirmed during a technical discovery or proof of concept.

03 | SERVICE SELECTION

What to use and when

The same cloud platform can support very different workloads. The table below is a practical selection guide for matching a requirement to the platform capability described in the source material.

If you need to...	Use this capability	Example
Run a conventional business application	Managed virtual machines, virtual networking, storage, firewall, and load balancing.	Three-tier application with web, application, and database servers separated into controlled network segments.
Modernize an existing application	OpenShift container platform with self-service provisioning and managed network/security services.	Move an application from fixed servers into containers, then scale application components independently as demand changes.
Build a cloud-native service	OpenShift for containers, APIs, and repeatable deployment patterns; use load balancing for ingress.	Deploy a customer-facing API with separate services for identity, orders, and notifications.
Extend an on-premises data center	Sovereign public-cloud resources with VPN, firewalling, identity controls, and managed connectivity.	Burst test environments or disaster-recovery capacity into the cloud while keeping core systems on premises.
Run large-scale model training or HPC	BareMetal servers with optional NVIDIA InfiniBand support, subject to availability and workload validation.	Distribute training jobs across GPU or accelerator nodes where high-throughput interconnects are material to performance.
Reduce platform operations overhead	Managed VMs, networking, backup, and infrastructure services backed by enterprise support.	Let a small IT team standardize backup and network operations without building every capability in-house.

Example deployment sequence

1. Define the workload boundary and data-handling requirements. 2. Select VM-based infrastructure or the OpenShift application platform. 3. Create network segments and access controls. 4. Deploy the application, service, or data tier. 5. Add backup, monitoring, and support processes. 6. Load-test the environment and tune capacity before production cutover.

04 | SECURITY AND SOVEREIGNTY

Controls that shape the design

Security is not a single product choice; it is a set of decisions that should be applied at the identity, network, workload, data, and operations layers.

Control area	Technical role	Design question
Identity	Multi-factor authentication and controlled administrator access reduce the risk of credential compromise.	Which users, service accounts, and support roles can access each environment, and how is access reviewed?
Network	Firewalls, segmentation, load balancing, and SSL-VPN establish controlled paths into and across workloads.	Which traffic is public, private, administrative, or service-to-service?
Data protection	Encryption and backup help protect data in transit, at rest, and during recovery operations.	What are the retention, recovery-point, recovery-time, and key-management requirements?
Sovereignty	The source material positions SpeedCloud as keeping data, operations, and technology within India.	Which residency, regulatory, and support-location requirements must the architecture satisfy?
Operations	Managed services and enterprise support can reduce operational load while formalizing ownership boundaries.	Which tasks remain with the customer, and which are covered by the managed service?

For regulated or business-critical workloads, validate the platform against the organization's required certifications, contractual controls, incident processes, audit evidence, data residency rules, and recovery objectives. Those details are not specified in the supplied material and should be confirmed directly with NxtGen during solution design.

05 | PERFORMANCE AND ECONOMICS

How to read the supplied benchmark claims

The provided product visual compares SpeedCloud with AWS, GCP, and Azure across compute, storage, transfer, and network-oriented metrics. It highlights directional claims rather than a complete independent benchmark report.

Claim shown in supplied visual	What it suggests	What to validate
Up to 179% performance boost	A potential advantage for selected single-core and multi-core compute tests.	Instance types, CPU generations, compiler settings, workload mix, and test duration.
Up to 1305% faster on average	A large claimed advantage for selected storage or I/O tests in the graphic.	Block or file storage configuration, queue depth, caching, dataset size, and measurement method.
Up to 6400 MT/s transfer rates	High transfer-rate capability for the measured data-transfer scenario.	Protocol, packet size, path, concurrency, source and target locations, and sustained versus burst rate.
Over 670% performance gains	A claimed advantage in selected network-transfer or bitrate tests.	Network topology, adapter type, sender/receiver configuration, and real application behavior.
Up to 50% lower TCO	A potential cost advantage versus leading cloud service providers.	Like-for-like instance, storage, egress, support, backup, licensing, and operating-cost assumptions.

Benchmark visual

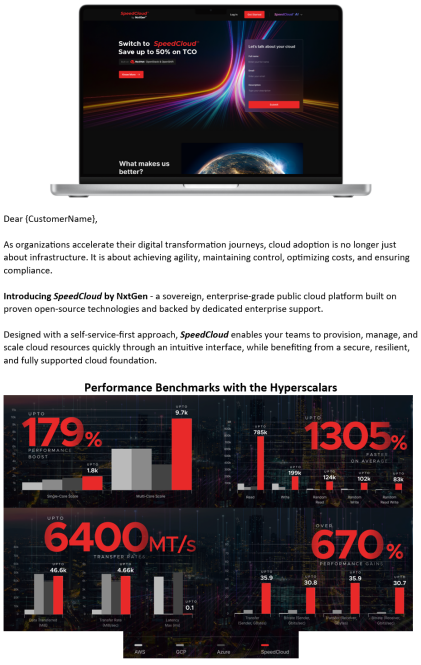


Figure 1. Product and benchmark graphic supplied with the EDM source material. The percentages and maxima are presented as marketing claims; confirm workload-specific results through a proof of concept.

06 | WHY TEAMS CHOOSE IT

Capability summary

The second supplied screenshot summarizes the platform's principal differentiators. The technical interpretation below turns each differentiator into an architectural or operating implication.

Supplied differentiator	Technical interpretation
Sovereign Cloud Assurance	Use residency and support-location requirements as design inputs for workload placement, access, and governance.
Enterprise-Backed Open Source	Use OpenStack and OpenShift to gain open-source foundations while using enterprise support to reduce platform risk.
Security at Every Layer	Combine identity, network, encryption, remote access, and workload controls rather than relying on a single perimeter.
Performance-Driven Infrastructure	Select compute, storage, networking, or bare metal based on the workload's actual bottleneck and performance objective.
Scale on Demand	Design for capacity changes without over-provisioning the initial environment; validate quotas and scaling limits.
Managed Services for Operational Simplicity	Use managed infrastructure and backup services to keep internal teams focused on applications and business outcomes.
High-Performance Network Capabilities	Consider high-throughput interconnects for training, HPC, and data-intensive workloads where network time materially affects total job time.

Important note

This PDF is a descriptive technical guide derived from the provided EDM source document and screenshots. Product availability, performance, certifications, service limits, pricing, and final architecture should be confirmed with NxtGen for the intended workload and geography.

07 | PRACTICAL EVALUATION

A focused discovery checklist

Use the questions below to convert the platform overview into a concrete design and commercial evaluation.

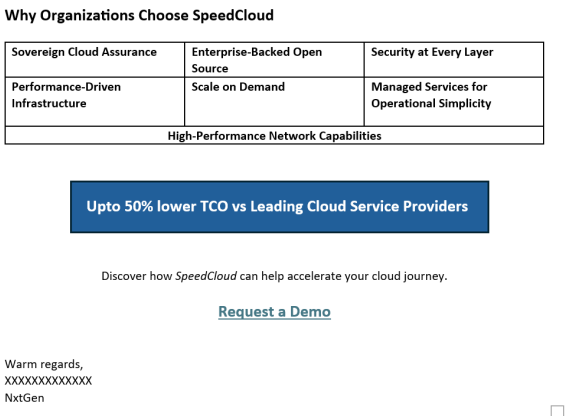
Area	Questions to answer
Workload profile	What applications, databases, containers, batch jobs, or AI/HPC workloads are in scope? What are the peak and steady-state resource requirements?
Data and compliance	Which data must remain in India? What retention, encryption, audit, and access requirements apply?
Network design	What needs public access, private access, VPN connectivity, load balancing, or service-to-service communication?
Operations	Who provisions resources, approves changes, responds to incidents, patches systems, and owns backups?
Resilience	What are the recovery-point and recovery-time objectives? How will restore tests be performed?
Performance	Which metrics matter: latency, IOPS, throughput, CPU time, job completion time, or application response time?
Economics	What is the complete cost baseline, including egress, support, backup, licensing, staff effort, and expected growth?

Recommended next step

Run a workload-specific proof of concept that includes one conventional VM workload, one containerized application, one security and connectivity path, and one backup-and-restore exercise. If AI or HPC is in scope, add a representative multi-node test that measures end-to-end job time, not only peak link rate.

The supplied collateral invites readers to request a demo. A useful demo should show the complete path from provisioning to secure application access, operational visibility, backup, scaling, and support handoff.

Source visual: capability summary



Why Organizations Choose SpeedCloud

Sovereign Cloud Assurance	Enterprise-Backed Open Source	Security at Every Layer
Performance-Driven Infrastructure	Scale on Demand	Managed Services for Operational Simplicity
High-Performance Network Capabilities		

Upto 50% lower TCO vs Leading Cloud Service Providers

Discover how *SpeedCloud* can help accelerate your cloud journey.

Request a Demo

Warm regards,
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NxtGen

Figure 2. Capability summary supplied with the EDM source material.